

WASILLA, AK, USA

WMO: 702325

Lat: 61.572N

Lon: 149.541W

Elev: 354

StdP: 14.51

Time Zone: -9.00 (NAK)

Period: 98-19

WBAN: 26443

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.1	-8.7	-26.8	1.2	-5.9	-24.0	1.5	-1.3	27.0	23.4	23.5	25.9	1.4	250	0.646

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.4	74.9	60.1	72.1	58.6	68.4	56.9	62.0	73.2	59.8	69.4	58.1	66.2	6.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
56.8	69.8	65.8	55.0	65.2	63.0	54.1	63.1	62.1	27.8	73.3	26.4	69.7	25.1	65.9	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.9	18.0	15.1	-17.5	79.2	8.5	5.0	-23.6	82.8	-28.6	85.7	-33.4	88.5	-39.6	92.2		
(5)			-18.1	64.5	7.7	3.2	-23.7	66.8	-28.1	68.7	-32.4	70.4	-38.0	72.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.7	17.6	22.2	27.0	38.2	48.0	55.5	58.6	56.5	48.4	37.0	23.7	18.7	
	DBStd	17.47	14.51	12.59	10.16	7.20	5.48	4.67	4.30	4.57	5.25	8.35	11.78	12.83	
	HDD50	5212	1003	778	714	355	101	6	0	3	86	405	789	972	
	HDD65	9974	1468	1198	1179	803	526	288	205	267	497	867	1239	1437	
	CDD50	725	0	0	0	2	40	170	267	204	39	3	0	0	
	CDD65	12	0	0	0	0	0	2	7	3	0	0	0	0	
	CDH74	160	0	0	0	0	3	40	79	38	0	0	0	0	
	CDH80	21	0	0	0	0	0	5	13	3	0	0	0	0	
Wind	WSAvg	4.9	5.2	5.4	6.9	5.4	4.6	3.8	3.4	3.5	4.1	5.5	5.6	5.5	
Precipitation	PrecAvg	22.2	1.3	1.3	0.9	0.5	1.0	1.3	2.1	3.4	4.2	2.3	1.5	1.6	
	PrecMax	30.9	2.8	2.8	1.9	1.6	1.9	2.8	4.0	8.3	12.5	4.9	4.1	5.2	
	PrecMin	16.6	0.2	0.2	0.3	0.0	0.3	0.4	0.5	1.3	1.7	0.7	0.0	0.2	
	PrecStd	4.2	0.8	0.8	0.5	0.4	0.4	0.7	0.9	2.1	2.5	1.3	1.1	1.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.9	45.9	52.0	63.4	72.8	79.3	81.3	78.8	65.6	57.2	47.9	45.2	
		MCWB	38.9	38.0	41.2	48.6	54.7	62.3	62.8	62.3	53.9	47.1	41.1	38.1	
	2%	DB	43.0	43.2	45.9	56.6	68.1	73.1	75.1	73.1	62.6	52.4	45.3	42.8	
		MCWB	36.5	36.2	37.7	43.9	52.2	58.6	61.4	60.3	52.2	44.4	38.8	36.7	
	5%	DB	39.2	39.5	43.2	53.6	63.7	70.2	71.9	69.8	59.7	49.9	42.7	39.0	
		MCWB	33.9	33.7	35.3	41.6	49.8	57.1	59.7	58.2	50.8	42.8	36.9	33.7	
	10%	DB	36.7	37.2	39.4	49.9	60.8	66.0	68.0	65.3	56.9	47.7	38.8	36.5	
		MCWB	31.5	31.6	32.9	39.7	47.8	55.0	58.2	56.2	49.4	41.6	34.1	31.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.0	39.4	42.0	49.2	56.4	63.8	66.2	64.1	55.5	48.4	41.4	39.1
			MCDB	45.4	44.4	49.6	62.3	71.1	76.5	79.5	75.4	62.7	56.1	46.4	43.5
2%		WB	37.3	37.2	39.0	45.0	52.8	60.0	62.3	61.3	53.7	45.7	39.2	37.0	
		MCDB	42.6	42.8	45.1	54.8	65.2	71.6	73.6	71.6	59.6	50.5	43.9	42.2	
5%		WB	34.7	35.0	36.2	42.1	50.7	57.5	60.2	59.1	52.3	43.8	37.3	34.5	
		MCDB	39.6	40.2	42.4	51.3	62.0	67.6	69.8	67.3	57.6	48.5	41.8	38.8	
10%		WB	32.0	32.4	33.8	40.2	48.9	55.5	58.6	57.2	50.8	42.0	35.0	31.7	
		MCDB	36.4	37.3	40.2	48.6	59.0	64.9	66.6	63.8	55.8	46.8	38.7	35.9	
Mean Daily Temperature Range	5% DB	MDBR	12.8	14.0	14.8	16.8	18.5	16.5	14.4	14.4	13.8	11.7	11.8	12.3	
		MCDBR	10.8	11.1	15.6	23.0	26.1	24.0	22.4	21.1	17.5	11.8	9.6	11.5	
		MCWBR	7.6	8.2	10.7	13.6	13.8	12.6	11.8	11.2	10.2	7.2	6.6	8.6	
	5% WB	MCDBR	10.2	10.7	15.1	20.9	23.5	21.7	19.8	18.4	15.1	10.8	9.0	11.4	
		MCWBR	7.4	8.2	10.6	12.7	12.7	12.0	11.0	10.5	9.5	7.1	6.4	8.9	
Clear-Sky Solar Irradiance	taub		0.226	0.278	0.298	0.336	0.358	0.358	0.364	0.348	0.310	0.290	0.237	0.377	
	taud		2.115	2.248	2.286	2.268	2.300	2.386	2.404	2.462	2.551	2.436	2.098	2.679	
	Ebn at Noon		158	222	260	271	273	275	270	266	259	220	145	95	
	Edh at Noon		11	20	27	34	36	34	32	28	22	17	11	8	
All-Sky Solar Radiation	RadAvg		88	311	828	1386	1649	1692	1509	1198	776	357	130	45	
	RadStd		12	36	76	108	129	108	115	120	98	40	18	6	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)		N/A	+3.56	+2.95	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page